Practice workbook · Describe phagocytosis, antibody production and barriers to pathogens

White-cell defence (engulfing and digesting, antibodies) and the physical and chemical barriers that keep pathogens out.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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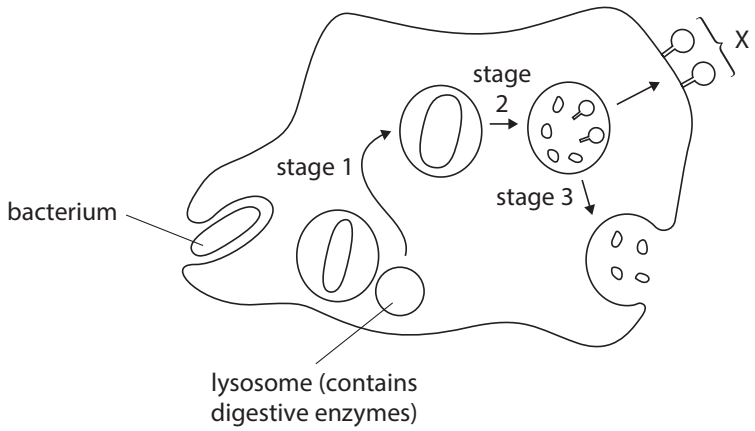
Type ID: HBI-DESCRIBE-PHAGOCYTOSIS-AND-BODY-DEFENCES

1. Question 6 · part a.ii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part a.ii only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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1. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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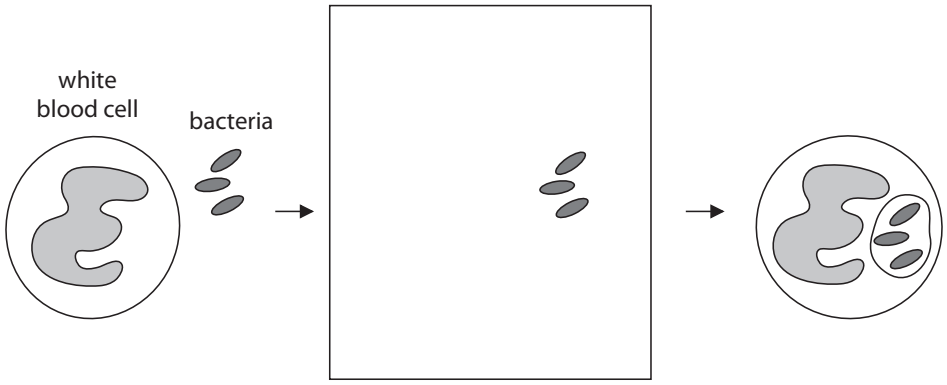
2. Question 2 · part a.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.i only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

(1)

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2. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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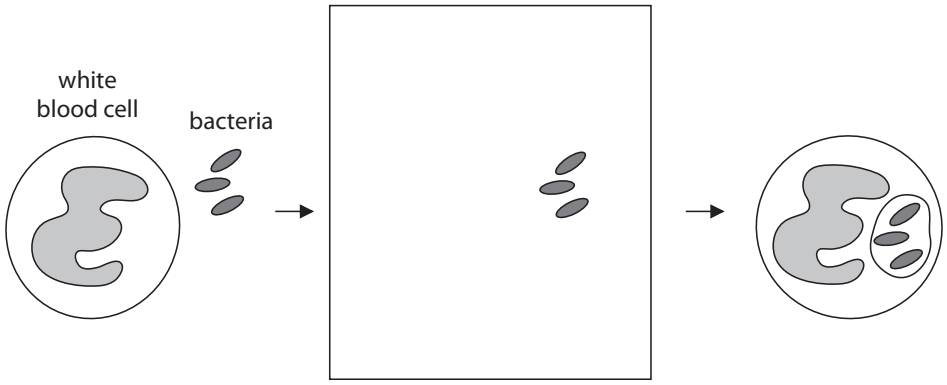
3. Question 2 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

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The diagram shows how one type of white blood cell helps to defend the body against disease.



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3. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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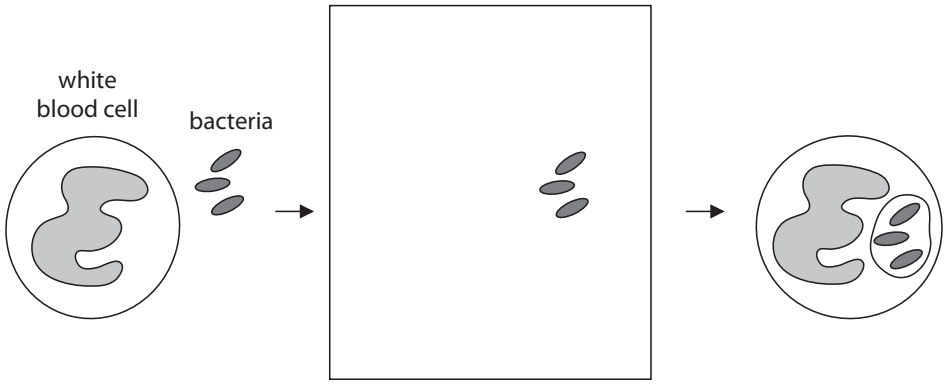
4. Question 2 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

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4. Question 2 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

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(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

(3)

Structures	Bacterial cell	Human skin cell
nucleus		
DNA		
cytoplasm	✓	✓
cell wall		

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(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

(1)

☒ **A** a double-stranded helix containing the bases ATGC

☒ **B** a double-stranded helix containing the bases AUGC

☒ **C** a single-stranded helix containing the bases ATGC

☒ **D** a single-stranded helix containing the bases AUGC

(Total for Question 2 = 8 marks)

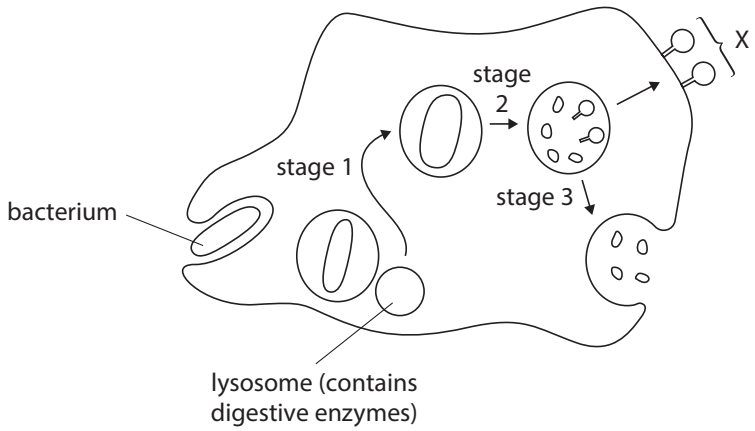
Working and answer

5. Question 6 · part a.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.i only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

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5. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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6. Question 8 · part a.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.i only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

.....

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

.....

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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6. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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7. Question 8 · part a.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

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8 (a) Read the passage below.

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Describe the process of phagocytosis. (lines 4 to 6)

(2)

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7. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 8 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

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(2)

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8. Question 8 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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8. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

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9. Question 4 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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9. Question 4 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1

2

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people =

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9. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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